

Supplementary methods

Synthetic-guided feature discovery in mouse spaceflight transcriptomics

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Frozen analysis supplement. This document records exact data roles, architecture, evaluation gates, statistical safeguards, output locations, and rebuild commands. It does not rerun training.

S1. Reproducibility contract

The manuscript builder consumes completed outputs and fails when a required file or a key expected result is missing. It does not:

- query the OSDR API;
- preprocess expression;
- train or fine-tune a neural network;
- sample a new synthetic cohort;
- rerun feature selection;
- recalculate random-effects statistics from raw profiles.

The exact frozen inputs and SHA-256 hashes are in `source_data/frozen_input_manifest.tsv`. Figure hashes are in `source_data/figure_build_manifest.tsv`.

S2. OSDR discovery and expression ingestion

OSDR data were obtained through the Biological Data API documented at <https://visualization.osdr.nasa.gov/biodata/api/>. The repository implementation and endpoint notes are in `docs/osdr_api.md`.

Eligibility required:

- `Mus musculus`;
- transcription profiling by bulk RNA sequencing;
- a resolvable spaceflight/flight or ground-control label;
- processed RSEM expected-count output;
- sample-level tissue or material metadata sufficient for canonicalization.

No raw combined OSDR HDF5 file was used. The API audit outputs are:

```
outputs/generative_benchmark/data_audit/osdr/osdr_canonical_metadata.tsv
outputs/generative_benchmark/data_audit/osdr/osdr_inventory_summary.json
outputs/generative_benchmark/data_audit/osdr/osdr_tissue_alias_audit.tsv
outputs/generative_benchmark/data_audit/osdr/osdr_tissue_inventory.tsv
```

The API returned 1,631 profile rows. Twenty-one technical replicates were aggregated, leaving 1,610 biological profiles, 835 flight and 775 ground control, from 75 accessions and 24 canonical material classes.

S3. ARCHS4 cohort audit

The source file was `assets/archs4/mouse_gene_v2.5.h5`, containing 997,515 profiles and 53,511 genes. The complete audit files are:

```
outputs/generative_benchmark/data_audit/archs4/archs4_full_profile_audit.tsv.gz
outputs/generative_benchmark/data_audit/archs4/archs4_control_only_balanced.tsv.gz
outputs/generative_benchmark/data_audit/archs4/archs4_healthy_preferred_balanced.tsv.gz
outputs/generative_benchmark/data_audit/archs4/archs4_broad_balanced.tsv.gz
```

The three eligible reference cohorts contained:

Cohort	Profiles	GEO series	Intended use
Control only	23,614	3,213	Conservative sensitivity cohort
Healthy preferred	62,299	5,307	Primary pretraining source
Broad	134,250	15,111	Diversity sensitivity cohort

The paper-parity run selected 17,244 healthy-preferred profiles across 20 tissue classes with whole-series assignment to 9,796 training, 2,448 validation, and 5,000 test profiles.

S4. Landmark panel and normalization

Full-transcriptome TPM was calculated with `data/reference/genencode_vM39_mouse_gene_lengths.tsv`. Landmark selection occurred after TPM calculation. The deterministic 974-gene panel and the human-to-mouse mapping audit are:

```
data/diffusion/l974_mouse_paper_parity.tsv
data/diffusion/l1000_human_to_mouse_ensembl.tsv
```

Training-set MaxAbs scaling was applied after landmark selection. The ARCHS4 prepared matrix is:

```
outputs/generative_benchmark/data/lacan_diffusion/archs4_mouse_paper_parity_l974.h5
```

The OSDR factorized matrix and profile metadata are:

```
outputs/generative_benchmark/data/lacan_diffusion/
osdr_factorized_within_study_replicated_validation_l974.h5
outputs/generative_benchmark/data/lacan_diffusion/
osdr_factorized_within_study_replicated_validation_l974.samples.tsv.gz
```

S5. ARCHS4 DDIM configuration

The full configuration is `configs/rna_diffusion/archs4_mouse_paper_parity.yaml`. The official source commit was `cde890154698fcea96c924804aaff04af3351b48`.

Component	Value
Input genes	974
Hidden layers	8,192; 8,192
Parameters	227,109,786
Dropout	0.1
Diffusion steps	1,000
Beta schedule	Quadratic, 0.0001 to 0.02
Objective	Summed noise MSE
Optimizer	Adam
Learning rate	0.0004783833151836702
Scheduler	OneCycle
Batch size	2,048
Epochs / optimizer steps	15,000 / 75,000
AMP	Enabled
EMA	0.999
Device	NVIDIA A100-SXM4-40GB
Runtime	5,987 seconds
Peak allocated GPU memory	5.93 GB

Run directory:

```
outputs/generative_benchmark/runs/lacan_diffusion/archs4_mouse_paper_parity_seed1234/
```

S6. Factorized OSDR adaptation

The accepted configuration is `configs/rna_diffusion/osdr_factorized_study_lora512_correlation_refine.yaml`.

Component	Value
Backbone	Completed ARCHS4 paper-parity DDIM
Conditioning	Tissue, FLT/GC, accession, material type
Domain adapter	LoRA rank 512, alpha 512
Domain stage	4,000 steps, LR 0.00002
Condition stage	1,000 steps, LR 0.00002
Batch size	512
Condition dropout	0.15
Correlation regularization	Weight 10, 256 genes, timestep ≤ 200
Sampling	100 DDIM steps for evaluation

The data split contained 781 training, 536 validation, and 293 locked-test profiles. Every test accession was represented in training. Results therefore measure within-study interpolation.

The accepted calibrator used:

1. train-only global alignment;

2. hierarchically shrunk accession and tissue means;
3. positive missing-covariance residual noise;
4. no condition-specific fit of calibrator means or covariance;
5. explicit clipping at zero for accepted downstream expression.

Run directory:

```
outputs/generative_benchmark/runs/lacan_diffusion/
osdr_factorized_study_lora512_correlation_refine_seed2020/
```

S7. Distribution and condition gates

Four generation seeds, 5020-5023, were declared before the locked test was opened. Every metric was gated independently.

Metric	Gate
Gene-correlation agreement	At least min(0.98, real-bootstrap P05)
Precision	≥ 0.95
Recall	≥ 0.85
F1	≥ 0.90
Adversarial accuracy	0.40 to 0.60
FD / real-split P95	≤ 1.0
Pooled effect recovery	Correlation ≥ 0.30 and direction ≥ 0.55
Muscle accession recovery	Correlation ≥ 0.30 and direction ≥ 0.55
Memorization	Generated fraction below train LOO P01 ≤ 0.05

The exact repeat rows are in `source_data/table_s2_locked_ddim_repeats.tsv`.

The term "adversarial accuracy" refers to an external nearest-neighbor real-versus-synthetic classifier, not the WGAN training critic. A result near 0.5 indicates that this external discriminator cannot reliably separate the two cohorts.

S8. WGAN-GP and GeneJEPa screens

The WGAN used the Viñas et al. topology: 64-dimensional noise, two 256-unit generator and critic layers, five critic updates, gradient-penalty weight 10, RMSProp learning rate 0.0005, and batch size 32. The strongest study-conditioned validation result had external adversarial accuracy 0.6362. Because no calibration variant jointly fixed adversarial accuracy and retained the correlation floor, and because accession-aware FLT/GC recovery failed, its locked test remained unopened.

```
outputs/generative_benchmark/runs/vinas_wgan_gp/osdr_matched_study_conditioned_seed2020/
```

The exact-architecture GeneJEPa duration screen used 4,096 genes and 43,744 replacement-sampled training exposures. It reached 0.703 held-out tissue balanced accuracy versus 0.839 from expression. It is representation-only and has no expression decoder.

```
outputs/generative_benchmark/runs/genejepa/
matrix_phase_0_genejepa_exact_mouse_one_epoch_f2e01cf1f130d5cb/
```

S9. Generated-feature workflow

Five arms were compared:

1. `real_only`;
2. `generated_only`;
3. `real_plus_generated`, equal total real and synthetic weight;
4. `guided_real_only`, real classifier with real/synthetic consensus ranking;
5. `guided_low_weight`, real plus recentered synthetic profiles at 0.05 total synthetic weight.

Splits were nested within accession-by-condition strata. The inner loop selected feature count, regularization, and rank method. The outer loop measured balanced accuracy, AUROC, and average precision separately. No composite score was used.

Stable genes had selection frequency at least 0.50 and coefficient-sign agreement at least 0.75. Generated-supported status did not constitute biological evidence. Real random-effects and LOO tests were applied afterward.

Primary workflow documentation:

```
docs/generated_feature_guidance_workflow.md
```

S10. Independent confirmation

The confirmation protocol is frozen at:

```
outputs/generative_benchmark/analyses/generated_feature_guidance_confirmation_v1/protocol.md
```

Test accessions:

Tissue	Accession	Profiles	FLT	GC
Lung	OSD-900	20	10	10
Thymus	OSD-457	24	12	12

Both accessions were removed from all generator-adaptation roles. The existing 15,000-epoch ARCHS4 checkpoint was reused, and only the 5,000-step OSDR adaptation was rerun. OSD-464 lung and OSD-244 thymus were fixed validation studies.

The deployed thymus classifier used real profiles only. Synthetic data changed feature ranking. The deployed lung classifier used a recentered synthetic view at 0.05 total sample weight.

Genotype assignment was audited after the primary result:

Tissue	Stratum	Profiles	FLT	GC
Lung	KO	10	5	5
Lung	WT	10	5	5
Thymus	Nrf2KO	12	6	6
Thymus	WT	12	6	6

S11. Random-effects and LOO rules

For gene (g) in accession (a), the real flight effect was:

```
delta[g,a] = mean(real expression[g] | FLT,a)
            - mean(real expression[g] | GC,a)
```

Accession effects were combined with a random-effects model. Benjamini-Hochberg FDR was calculated within each declared tissue/gene family. The LOO analysis removed each accession, repeated the random-effects fit, and retained the maximum FDR and any sign reversal.

A strict stable gene required:

- full-data random-effects FDR < 0.05;
- maximum LOO FDR < 0.05;
- no LOO sign reversal;
- a generated-informed selection status;
- agreement between the real effect and generated direction.

Generated profiles were never included in the random-effects model.

S12. Reactome analysis

The official mouse GMT is:

```
data/pathways/reactome_current_mouse_ensembl.gmt
```

It was generated from official `ReactomePathways.txt` and `Ensembl2Reactome_All_Levels.txt`, restricted to *Mus musculus*, R-MMU-* pathways, and ENSMUSG* genes.

Hypergeometric enrichment used the 974-gene landmark panel as background. Benjamini-Hochberg FDR was applied separately by tissue and selected-gene set. Reactome parent and child terms overlap. Counts of significant rows are therefore not counts of independent biological discoveries.

S13. Skeletal-muscle group analysis

The fixed DDIM and three frozen synthetic development views were reused. No neural network was retrained.

Group	Profiles	Accessions	FLT	GC
EDL	24	2	12	12
Gastrocnemius	25	3	10	15
Quadriceps	35	4	18	17
Soleus	41	3	22	19
Tibialis anterior	24	2	12	12

The full report is `docs/synthetic_skeletal_muscle_group_analysis.md`. Key frozen outputs are:

```
outputs/generative_benchmark/analyses/within_study_generated_feature_stability_muscle_groups_v1/
```

The seven soleus genes in the manuscript are the intersection of synthetic-selected genes, real-effect direction support, and strict real LOO FDR: `Bdh1`, `Bnip3`, `Mef2c`, `Ech1`, `Pxmp2`, `Gmnn`, and `Tpm1`.

LOO here is a real-data meta-analysis sensitivity test. It does not remove the accession from the already completed generator adaptation. Soleus remains developmental until a new accession is excluded from adaptation and selection.

S14. Supplementary figures

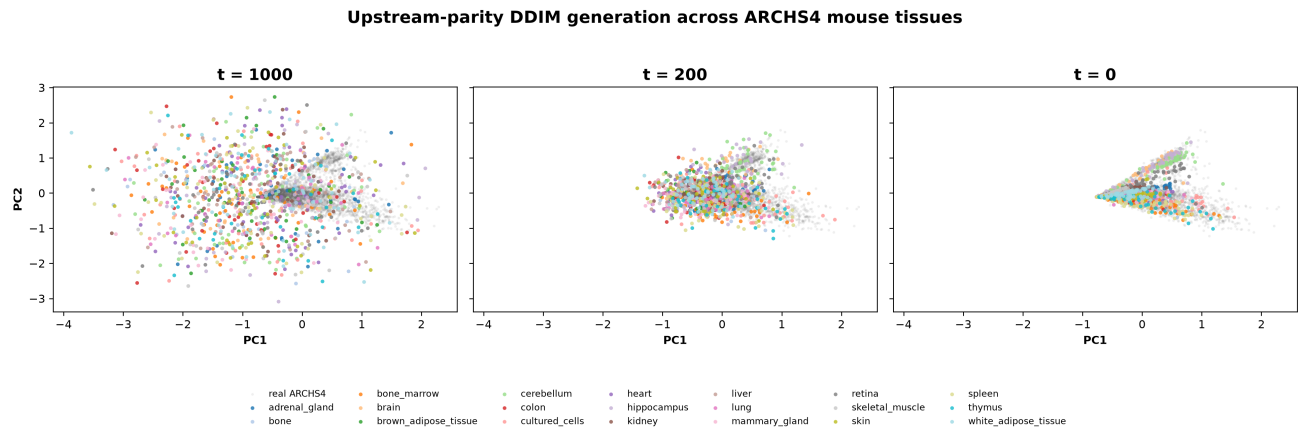


Figure S1. ARCHS4 DDIM denoising trajectory. The same generated profiles are shown at diffusion timesteps 1,000, 200, and 0 in a PCA space fitted to real ARCHS4 expression. Colors identify tissue classes. The two-dimensional view is descriptive; held-out tissue classification uses the full 974-gene representation.

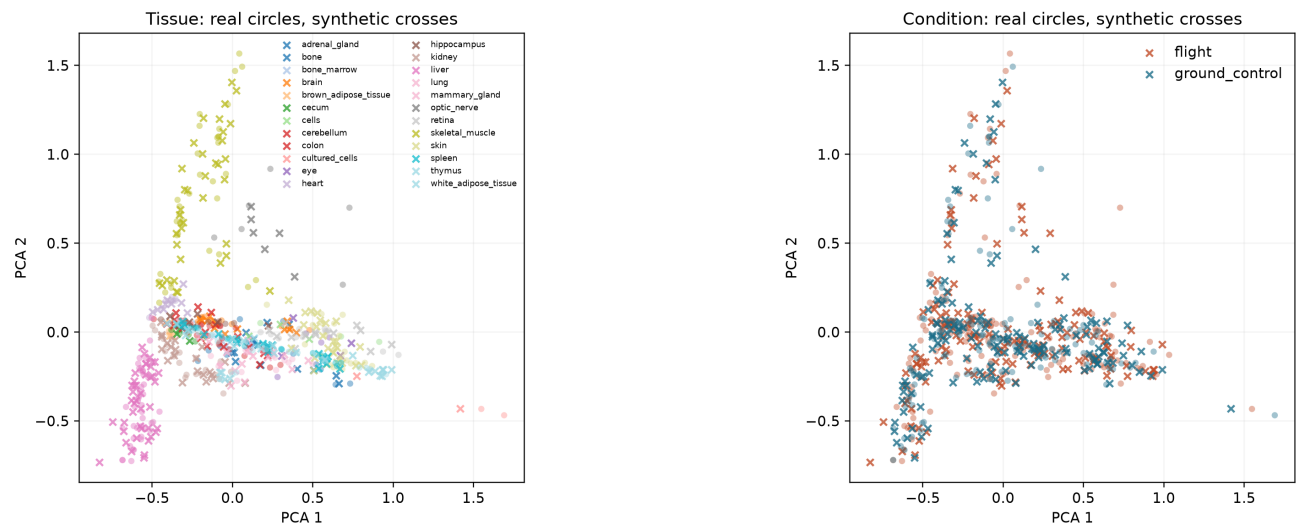


Figure S2. Real and generated profiles in the locked OSDR test. Seed 5020 is shown. Tissue and condition views are descriptive; formal fidelity and effect metrics use all declared seeds and higher-dimensional data.

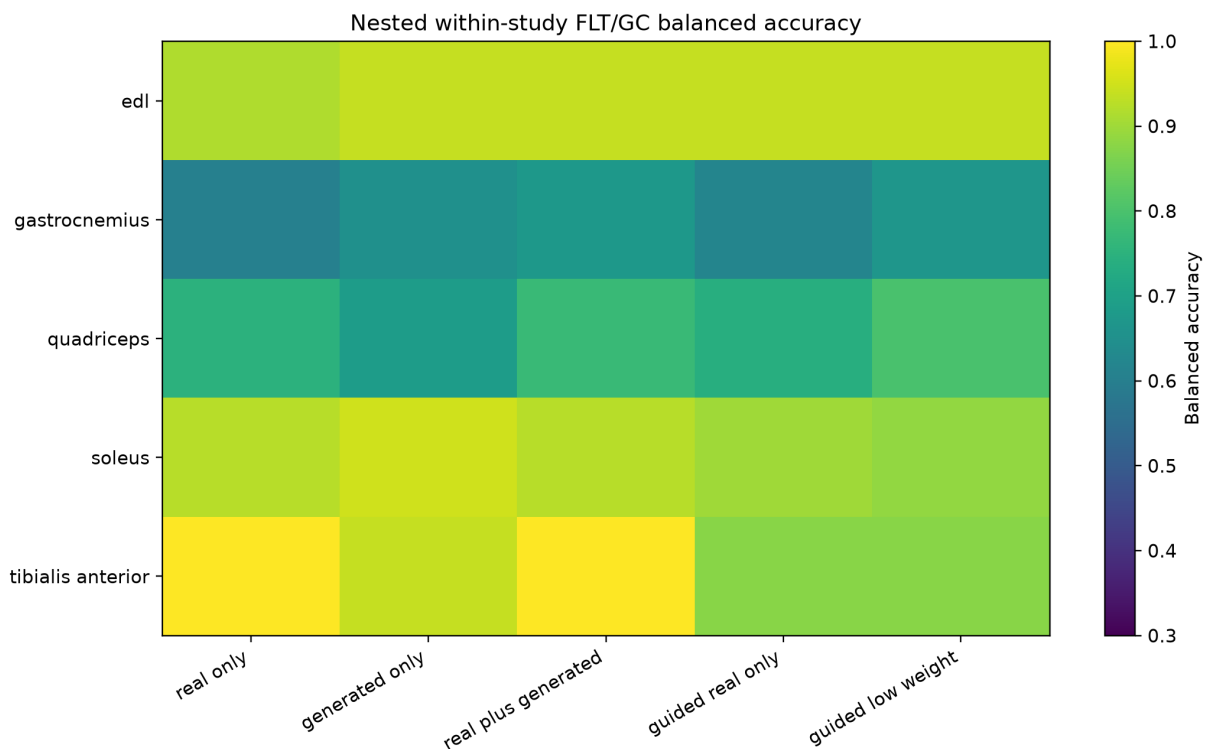


Figure S3. Repeated nested muscle-group balanced accuracy. Each row is a muscle group and each column is a downstream use of real or generated profiles. Arm selection also required nonworse AUROC and average precision.

S15. Source tables

- `table_1_data_inventory.tsv`
- `table_2_model_screen.tsv`
- `table_3_locked_ddim_metrics.tsv`
- `table_4_independent_confirmation.tsv`
- `table_5_tissue_evidence.tsv`
- `table_s1_archs4_ddim_metrics.tsv`
- `table_s2_locked_ddim_repeats.tsv`
- `table_s3_naive_augmentation.tsv`
- `table_s4_confirmation_genotypes.tsv`
- `table_s5_thymus_core_genes.tsv`
- `table_s6_thymus_reactome.tsv`
- `table_s7_muscle_group_summary.tsv`
- `table_s8_soleus_genes.tsv`
- `table_s9_muscle_reactome.tsv`
- `table_s10_all_tissue_development_screen.tsv`

S16. Rebuild command

```
PYTHONPATH=src /home/exouser/miniforge3/envs/nasa-mouse/bin/python \
-m nasa_mouse_rna_diffusion.build_synthetic_guided_paper
```

To regenerate source tables and figures without rendering PDFs:

```
PYTHONPATH=src /home/exouser/miniforge3/envs/nasa-mouse/bin/python \  
-m nasa_mouse_rna_diffusion.build_synthetic_guided_paper --skip-render
```